

A Comparison of Performance and Preference on Mobile Devices vs. Desktop Computers

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Abstract—Smartphones and desktop computers are ubiquitous in today's world. Due to the improved computing power and high portability of smartphone devices, users are performing more sophisticated tasks on their smartphones. This study compares the performance of a web-based task on a desktop computer with a mobile-based task on a smartphone. In addition, we have compared the overall user experience and preference across the two devices. In order to achieve this, we created a mobile-based word game for an Android mobile platform and a web-based version of the same word game for a browser platform. Sixty participants were instructed to complete the tasks on both devices. Our results show that while performance on the computer-based game was significantly better than performance on the mobile-based game, people preferred to play on the smartphone rather than the desktop computer. People preferred smartphones over desktop computers on account of their touch-screen features, portability and ease of use, even though they actually performed the task more efficiently on the desktop computers.

Keywords— *Smartphone; Mobile; Computer; Performance; Preference; User Experience*

I. INTRODUCTION

People own and operate multiple computing devices of different screen sizes. The usage of smartphones and desktop computers is dependent on the type of tasks performed. Due to their smaller screen size, smartphone devices are primarily used for certain types of tasks like communications, managing calendars, and emailing. Smartphone screen sizes are smaller and therefore make it harder to read and understand large amounts of information. However, due to their high portability and continuous connectivity, smartphones are increasingly replacing desktop computers for work tasks [1]. Prior studies suggest that people use their smartphone devices differently than their desktop computers. For example, people will not edit long documents on their smartphone due to limited screen sizes and input facility [2]. Smaller screens make reading difficult and slower [3].

Mobile app-based games have increased in popularity in recent years. The prevalence of wireless network services and variety of phone sensors has motivated users to play games on smartphones frequently. In this study we examine people's device preferences as well as their performance when using smartphones and desktop computers. While several studies have examined task performance on desktop

computers and smartphones, to our knowledge they do not address both preference and performance in a single study.

II. BACKGROUND LITERATURE

Prior research suggests that larger screen sizes positively impact task performance. According to Bao, Pierce, Whittaker, and Zhai [2], smartphones were used for viewing information rather than producing information. Research done on non-mobile business users revealed that high performances are achieved for reading tasks and lower performance for typing tasks on smartphones. Jones, Buchanan, and Thimbleby [4] observed that participants using small screens were 50% less effective in completing tasks than the large screen participants. It was difficult for participants to make a good decision about the search results obtained on smaller devices, resulting in lower performance. Marcial and Hemminger [5] suggest that as screen size is reduced from desktops to tablets and yet further to smartphones, task execution time increased significantly. The within-document searching task took longer on phones resulting in lower performance. This supports the results of Jones, Buchanan, and Thimbleby [4] that web searches are more difficult on smaller screens compared to larger screens.

Raptis, Tselios, Kjeldskov, and Skov [6] examined the effect of screen size on Internet browsing using three different devices of the same brand with varying screen sizes (3.5in, 4.3in, and 5.3in). They observed that participants performed more efficiently on devices with a screen size larger than 4.3in. They also suggested that not all tasks are affected by the screen size. Tasks which were difficult to complete and require scrolling showed significant differences in efficiency. Kim and Albers [7] observed that search operation on smaller display screens took longer, however no significant difference in performance was found for participants using a smaller device compared to those using a desktop computer. Participants, however, expressed frustration at performing search tasks with PDA devices and said they would not use it in this way.

According to Karlson, Meyers, Jacobs, Johns, and Kane [8], portability and network connectivity influence the usage of smartphones compared to desktop computers. They observed that people prefer smartphones to check their emails and manage calendars as it requires no boot up and password entry is fast. Karlson et al. [1] found that people do

not rely on mobile devices for composing long emails, but for checking emails, deleting emails, and composing short emails.

According to Sanchez and Branaghan [9] there was no significance difference between learning and recalling information on smaller and larger screen devices. However, when users were asked to produce reasoning or make decisions based on the information learned, their performance decreased on smaller devices. This supports Bao, Pierce, Whittaker, and Zhai's [2] result that information producing tasks' efficiency on smaller screens is significantly lower compared to that on larger display sizes. Sanchez and Branaghan [9] imply that reducing scrolling interfaces on smartphones may increase performance.

Studies have shown that smartphone users have fragmented attention compared to desktop or laptop computer users. According to Oulasvirta, Tamminen, Roto, and Kuorelahti [10], smaller screens prevented users from performing more complex tasks. Couper and Peterson [11] found that surveys on smartphones took longer to complete than surveys on PCs. Similarly, Reeves, Lang, Kim, and Tatar [12] found that people pay more attention to larger displays sizes and were more aroused compared to smaller screens.

The overall goal of this study is to examine whether device screen size effects the game performance and users preference. To achieve this, the following research questions were addressed:

Research Questions:

R1 Which device do people prefer to play games on: desktop computer or smartphone?

R2 Why do people prefer a desktop computer or smartphone?

R3 Which device do people perform better on?

III. HYPOTHESES

Previous research has shown that people performing tasks on devices with larger screen sizes will have a higher performance compared to those performing the same tasks on devices with smaller screen sizes [2, 5, 6]. Therefore, we hypothesize:

H1. People will perform better on a desktop computer than a smartphone.

As smartphone penetration continues to grow, people prefer it to desktop/laptop computers due to its high portability and ease of use [1, 8]. Therefore we hypothesize:

H2. People will prefer to play on a smartphone rather than a computer.

IV. METHODOLOGY

In order to examine the differences in performance and preference on smartphones and computers, we developed a

mobile-based and web-based version of the same word game.

A. Participants

We recruited 60 undergraduate and graduate students (27 female and 33 male) from a university in the Midwest. All participants had to have prior experience with smartphones. Each received a \$10 gift card to Best Buy or Target for their participation.

B. Procedure

Upon arriving in the lab, participants were requested to sign a consent form and were logged onto a desktop computer to begin the experiment. Each participant was given a smartphone as well. Participants answered demographic questions and questions about their typical desktop/laptop computer and smartphone usage. Next, they were given instructions for the game. In order to remove any learning effects, half of our participants began with the smartphone device while the other half started with the desktop computer. Participants were randomly assigned to one of these two groups. Participants were given 5 minutes to complete each task. When the 5 minutes elapsed participants filled out a post-test questionnaire indicating their overall experience with the specific device. The same process was repeated for the other device. Finally, after the second task was completed participants were directed to a post-experiment questionnaire with questions related to their device preference and brief explanation on their overall experience on two devices (smartphone and desktop computer).

C. Design

1) Within Subject Design/Counterbalanced repeated measure design

A within-subjects design was used. We conducted a complete counterbalanced repeated measure lab experiment.

All participants completed the task on both devices. As mentioned above, we randomly assigned participants in two groups: those who complete the web-based task on a desktop computer first and then complete the mobile-based task on a smartphone and those who complete the smartphone-based task first and then complete the web-based task on a desktop computer (see Fig. 1).

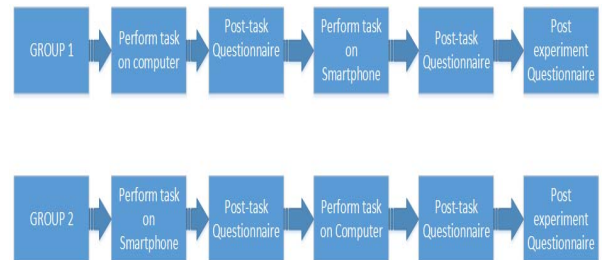


Fig. 1. Experimental design after completing the pre-experiment questionnaire

2) Application Selection

We selected Hangman as our test application. Hangman is a popular word guessing game. Hangman is an easy game to learn and play and requires fast thinking capability, which adds a lively activity with an extensive variability of words.

The goal is to guess the word in six chances. The word to guess is represented by a row of dashes indicating the number of letters in a word. If a participant selects the correct letter, a dash is replaced by a letter. Each wrong guess will draw a segment of Hangman in the gallows. The game ends either if the word is guessed correctly or if the word is not guessed in six chances.

3) Mobile-Based Game

For the mobile based task we created a Hangman application for Android platforms. The front end was created using Android SDK. See Fig. 2 for a screenshot of the Hangman mobile application.

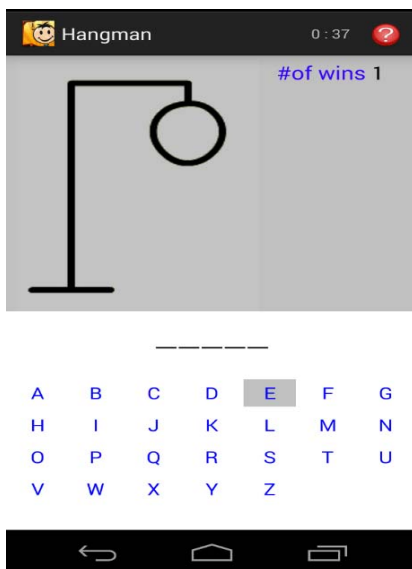


Fig. 2. Screenshot of Hangman Mobile App

4) Web-based Game

For the web-based game we created a Hangman application for HTML Web platform. See Fig. 3 for a screenshot of the Hangman Web application.

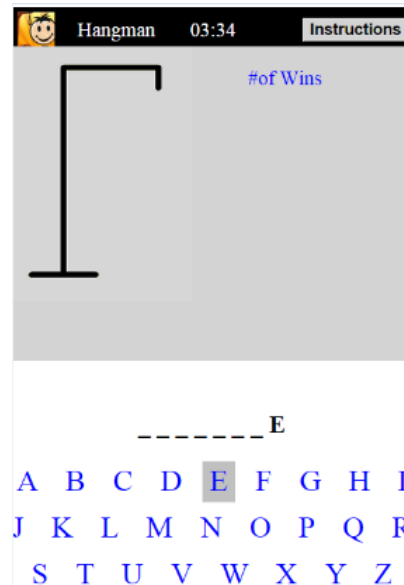


Fig. 3. Screenshot of the Hangman Web Application

D. Experiment

After completing the pre-experiment questionnaire, participants were randomly assigned to one of the two groups. Instructions for playing the Hangman web game were provided along with screenshots. Participants in group 1 (Computer first) were given five minutes to play the Hangman web-based game on a computer. Along with the game interface, participants were able to see their number of wins and a timer showing the remaining time. Whenever users checked the built-in instructions, the timer was paused to ensure everyone would spend 5 minutes on the game task. The browser back button was disabled to force the user to stay on the current task. When five minutes elapsed, participants were required to answer the post-task questionnaire for the web-based computer task.

Upon completion, group 1 participants were directed to the Hangman mobile-based game instructions page. Next, participants were asked to proceed to their assigned smartphone where the mobile app was opened up in full screen. Participants were given five minutes to play the game as many times as possible. Just as in the desktop environment, whenever the user checked the built-in instructions, the timer was paused. Furthermore, if the user attempted to go back or pressed home or the recent apps buttons, a pop-up warning would appear to force the user to stay on the current task. Upon completion, participants were asked to answer the post-task questionnaire for the mobile-based task. Finally, after completing both the web-based game and the mobile-based game, participants were directed to the post-experiment questionnaire. This process was reversed for group 2 (Smartphone first).

E. Questionnaires

1) Pre-experiment Questionnaire

Before beginning the experiment, we asked participants a series of questions pertaining to their demographics, such as age, gender, ethnicity, and English level proficiency. Additional questions were included to get user perception towards the computer and smartphone device like device preference for gaming, frequency of playing game, and their previous experience with the Hangman game application, smartphone and desktop/laptop computer activities.

2) Post-Web/Mobile-based Task Questionnaire

After completing each task on either the computer or smartphone, we asked participants to describe their overall task experience in one word.

3) Post-experiment Questionnaire

The post-experiment questionnaire comprised of questions documenting their device preference and explanation of why they prefer the particular device.

F. Measurement

During the experiment the following data were collected

Computer game wins: The average computer game scores. This was calculated by dividing number of game wins by the number of games played.

Smartphone game wins: The average smartphone game scores. This was calculated by dividing number of game wins by the number of games played.

Computer experience: One word description of the web-based game experience.

Smartphone experience: One word description of the mobile-based game experience.

V. RESULTS

In our analysis we examined the effects on game performance and user preference of the mobile-based task and web-based task on smartphones and desktop computers. We used repeated measure analysis of variance to test our hypothesis. We tested for any order effects which were found to be not significant, eliminating concern of learning or order effects between groups ($F(1, 58) = 0.034, p = 0.8536$).

A. Participants

The participants' age ranged from 18-70 years with a mean age of 25.58 ($SD=8.168$). There were 18 Caucasian, 9 Hispanic/ Latino, 5 Middle Eastern, 19 Asian Pacific Islander, 8 African American, and 1 other. The mean for English level was 4.68 (where 1 = Poor and 5 = Excellent), and the minimum was a 3.

B. Smartphone vs. Computer usage

Along with demographic questions the participants were asked to rate the top three usages of their smartphone and desktop/laptop computers in day-to-day life. For

smartphones the most common activities were Texting and talking (23%), Social Networking (16%) and Browsing (14%). See Fig. 4 for an activity summary on smartphones.

For desktop or laptop computers the most selected activities were School/Work activity (28%), Browsing (22%), Music/Movies (10%). See Fig. 5 for activity summary on desktop/ laptop computers.

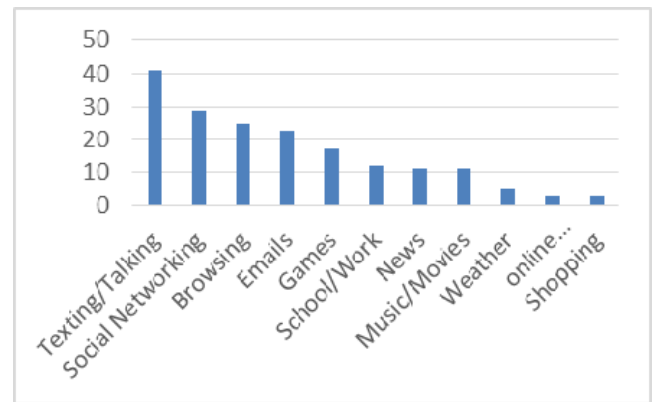


Fig. 4. Activity summary on Smartphone

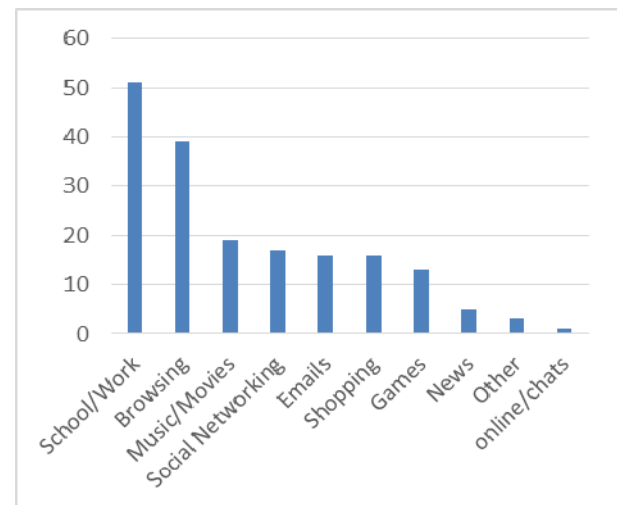


Fig. 5. Activity Summary on Desktop/Laptop Computer

C. Test of Hypothesis 1

To test whether participants would perform better on a desktop than a smartphone, we conducted within-participants t-tests to compare performance for the Hangman game, played in smartphone and desktop computer device. The game scores on the computer ($M=0.283, SD=0.187$) were significantly higher than on the smartphones ($M=0.231, SD=0.198$). This difference in performances on both devices was statistically significant (paired $t=2.15, p=0.0354$). This shows that participants performed significantly better on desktop computers compared to smartphones, supporting hypothesis 1. See Fig. 6 for performance differences in the smartphone-based and computer-based versions.

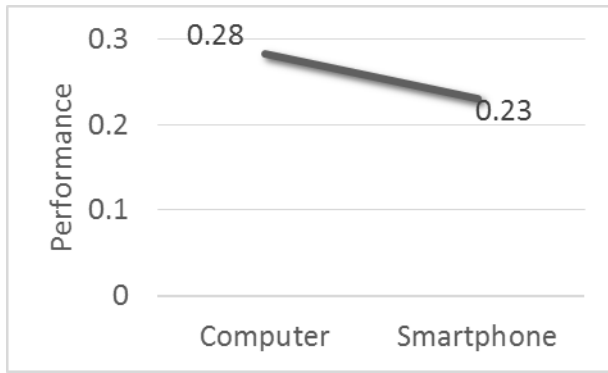


Fig. 6. Mean performance on smartphone mobile-based game and desktop computer web-based game

D. Test of Hypothesis 2

To test whether people would prefer to play on a smartphone device rather than a computer, as part of the post-task questionnaire we asked participants to describe in one word their overall experience playing the game on the desktop computer as well as on the smartphone device.

Two coders categorized the word into positive or negative. Cohen's Kappa Statistics yielded a reliability of 0.83 indicating a reliable measure. Chi-square tests were performed to determine the participants' perception of the game on both devices. Forty participants responded positively towards the mobile-based task on the smartphone compared to 18 who responded negatively. There was a significant difference found in smartphone experience, $\chi^2(1, N = 58) = 8.3448$, $p = 0.0039$. Thirty three participants responded positively toward the desktop-based task while 23 responded negatively. There were no significant differences for the browser-based desktop game, $\chi^2(1, N = 56) = 1.7857$, $p = 0.1814$. See Fig. 7 for the overall experience on the smartphone and desktop computers. Please note that those classified as neutral or other were removed from the analysis. Hence, hypothesis 2 was supported showing that participants preferred smartphones for playing games over a desktop computer.

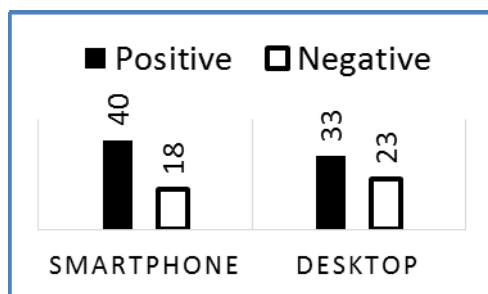


Fig. 7. Participant Experience on Smartphones and Computers

As part of the post-experiment questionnaire we asked participants which device they would prefer to play Hangman on. The smartphone device was ranked highest with 30 preferring it, compared to only 11 who preferred the desktop, showing a significant difference in user preferences $\chi^2(1, N=41) = 8.8049$, $p = 0.0030$. We excluded participants who preferred both devices ($N=19$). See Fig. 8 for device preference.

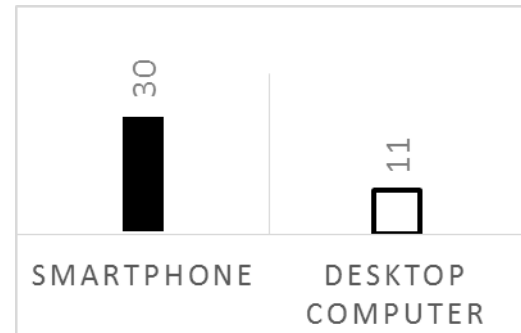


Fig. 8. Device Preference

Participant's reasons for preferring a smartphone suggested that touchscreen, portability and ease of use influenced their opinion. Some of the comments follow:

"It looks easier on the phone and more the touch screen made more interactive".

"I prefer touch screen devices easier and faster to use".

Participant's comments for preferring the computer implied that screen size, controllers (keyboard and mouse), and more computing power played a role in why they would prefer desktop computers over smartphones. Some of the comments were:

"I prefer to play with a mouse and keyboard as opposed to touch screen controls".

"Large screen, I think the computer screen's size catches my attention".

"It's better to visualize on a larger screen".

E. Between subject analysis

We were also interested in analyzing the performance in between subject's factor effects. This was determined by conducting a one-way ANOVA on the performance (difference between computer game scores and smartphone game scores) in two groups. A significant main effect was found ($F(1, 59) = 7.59$, $p = 0.0078$). See Fig. 9 for mean performance in the two groups.

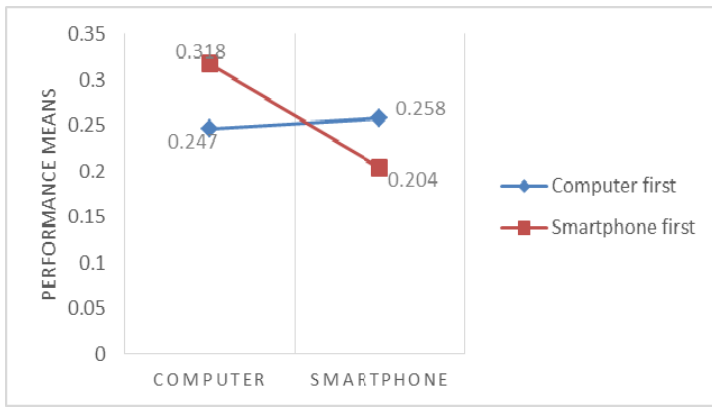


Fig. 9. Mean performance of tasks in two groups

The participants who started first on smartphone scored significantly higher in web-based game on desktop computer (Computer Mean = 0.318 vs. Smartphone Mean=0.204).

VI. DISCUSSION

Our study has provided evidence that device type has a significant effect on game performance. We observed that participants performed the web-based task more efficiently compared to the mobile-based task. This supports previous research which shows that people pay more attention when using larger screen sizes and perform much better than on smaller screen size devices [9]. The results also seem to support previous findings showing that people are faster using larger screen devices compared to smaller screen devices [3].

Analysis on subjective measures revealed that the participants preferred smartphones over desktop computers on account of the touchscreen, portability, and ease of use mobile characteristics. We found a significant difference in participants' smartphone experience. The majority of participants rated smartphones with significantly more positives than negatives, which was not true for desktop computers. This result seems to support previous findings that portability and ease of use highly influence smartphone usage [8].

This research is novel in that we examined performance and preference in a single study. We found that while users performed better on the desktop computers they preferred smartphones to play the word game. This implies that usability plays a major role in device preference as compared to performance.

This study has demonstrated that screen size impacts task performance. Larger screen sizes produce higher task efficiency compared to smaller screens. However not all tasks may benefit from larger screen displays [6]. Future work should examine performance and preference across devices in other tasks.

6.1. Limitations

This study is limited to a word game task. Further work is necessary to understand how our results will apply to different types of tasks. Also this study used college students as participants and therefore we suggest caution before generalizing the results to other populations.

VII. CONCLUSION

Smartphones are becoming part of our daily routine. Prior research suggests that high computing power and portability factor influence the smartphone usage. The goal of our study was to investigate both user performance and preference across different display conditions. In order to achieve this we conducted a within-subject design laboratory experiment. We developed the same task on mobile and web platforms. Each participant performed the mobile and web-based task under counterbalanced conditions to avoid learning effects.

We found that participants performed significantly better on desktop computers compared to mobile-based tasks. This suggests that larger screen sizes may have positive effects on task performance. We also found that though performance is better on desktop computers, participants preferred the smartphones. This was on account of the touchscreen, portability and ease of use of mobile devices.

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